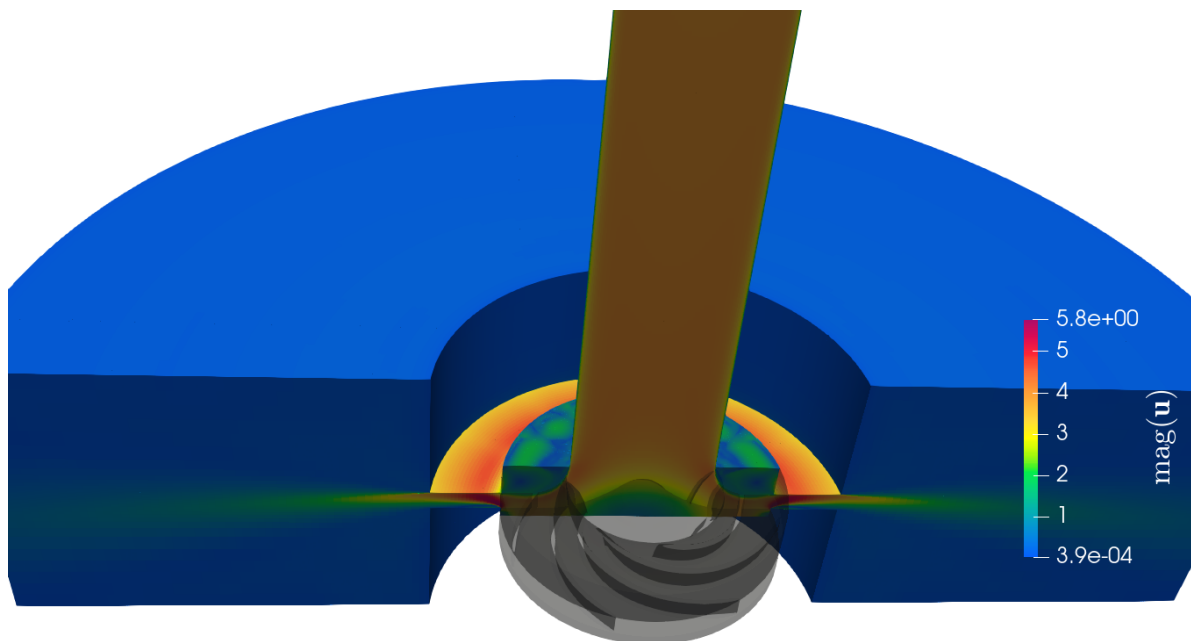


End-of-study thesis for engineering degree

Optimization of the performance of a centrifugal pump using Data Assimilation

Intern in Rotating Machines



Cross-section of the pump model

Educational tutor: Samy LABSIR

Laboratory tutors: Marcello MELDI, Antoine DAZIN, Miguel MARTINEZ VALERO

Clément THIBAULT - Aéro 5

From February 26th, 2025 to September 2nd, 2025

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Summary sheet

The aim of this technical assessment is to summarize the gaps and their causes between the course's missions and what has been achieved. It also enables us to suggest what could be done by the next person working on the same subject.

Summary sheet		Clément THIBAULT - Aéro 5 TIE	
Internship topic		Objectives	
Optimization of the performance of a centrifugal pump using Data Assimilation		Optimize the forcing parameters of an IBM ¹ in a centrifugal pump using DA with CONES	
Main customer		Used tools	
- LMFL - ENSAM		OpenFoam, CONES, VSCode, Git, Python, C++, Bash, TGCC, Paraview	
Conducted studies			
- Coding the DA problem in OpenFoam and CONES - Convergence of the CFD solution - Convergence of the EnFK			
Results		Explanation of possible differences	
- Diminution of 50% of the mathematical NRMSE with DA (almost 100% expected) - Implementation of inflation in CONES - Discovering of a recirculation bubble in the diffuser		- The shape of the rotating impeller combined with the IBM permit no-physical flow that induce recirculation bubble in the diffuser where DA sensors are located.	
Encountered difficulties		Work to be continued	
- Getting to grips with the tools OpenFoam and CONES - Code everything directly on the TGCC terminal - Initialized physical field of complex instationary simulation		- Use the same type of DA for the analysis of axial compressor internal flows - Synchronize PIV and simulation in the DA - Implemente interpolation IBM method	

Note: Acronyms are define later.

¹All acronyms are defined from the next page or available in the glossary.

General Introduction

Since I'm young, I've always been fascinated by fluid mechanics. I remember watching planes flying and being amazed by the way they could fly. I was also interested by the way water flows in rivers and sailing boats. In high school I continued more theoretically my planes projects by using Computational Fluid Dynamics (CFD). This fascination led me to study physics and mathematics in a university level, where I discovered applied mathematics including CFD, Artificial Intelligence (AI) and Kalman Filter. During my studies, I've become passionate about applied mathematics and fluid mechanics. That's why I did my last internship at Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique (CERFACS). It convinced me to do a thesis in CFD after Institut Polytechnique des Sciences Avancées (IPSA), my current engineering school. The subject proposed by Laboratoire de Mécanique des Fluides de Lille (LMFL) really suited me: CFD on an open-source library, coupled with AI and experimental data. So it was with enthusiasm that I started my internship on February 26, 2025, in this laboratory for a duration of six months.

In a few words, the LMFL is a public laboratory, composed by 38 researchers and around 25 PhD students. The laboratory has three sites in Lille (École Nationale Supérieure d'Arts et Métiers (ENSAM) and Office National d'Études et de Recherche Aéronautiques (ONERA)) and on the Villeneuve d'Ascq campus (Centrale Lille, Centre National de la Recherche Scientifique (CNRS), Université de Lille).

My main tasks during this internship were as follows:

- Convergence of the CFD solution;
- Implementation of the Data Assimilation (DA) problem in OpenFoam and Coupling OpenFoam with Numerical Environments (CONES);
- Optimization of forcing parameters of the Immersed Boundary Methods (IBM) using DA with CONES to correct a non-physical behavior using IBM;

To explain the content of my internship in more detail, I will first present the LMFL research laboratory. In the second part, I'll present the work I carried out in the form of a report. This second part is divided into five sub-sections:

1. Introduction to the DA problems;
2. Numerical ingredients of the CFD and DA problems;
3. IBM preliminary results;
4. Application of the DA to the IBM;
5. Results and perspectives.

Chapter 1

The LMFL Laboratory

1.1 Presentation of the LMFL Laboratory and Its Missions -or- Scientific Environment and Host Institution

The LMFL is a public laboratory under supervision of Centrale Lille, Université Lille 1, Arts et Métiers Paris Tech, ONERA and CNRS.



Figure 1.1: LMFL facilities; Top: Villeneuve d'Ascq's campus gathering Centrale Lille, CNRS and Université de Lille. Bottom-left: ONERA. Bottom-right: ENSAM (source : lmfl.cnrs.fr)

During the last six months, I have been hosted by ENSAM, more precisely at Lille campus. Arts et Métiers is known to be a prestigious engineering school with a focus on mechanics, industrial and energy engineering. Its Lille campus offers students unique opportunities to study in a dynamic and innovative environment, located in the heart of a region known for its industrial history.

This research structure was created on January 1, 2018 by the merger of two research units: the “Écoulements tournants et turbulents” team at the Lille Mechanics Laboratory, and the “Expérimentation et Limite de Vol” research unit.

This diversity of skills allows the research teams to be involved in numerous partnership projects in connection with regional or national structures.

Research conducted within the LMFL is organized into three main topics:

- **Turbulence** : Studying and modeling turbulent flows using numerical and experimental methods. A focus is made on inhomogeneity and non-stationarity in wall-bounded flows, wakes, jets and several other flow configurations. This research theme gathers 5 permanent researchers and about 10 PhD and post-doctoral students.
- **Rotating flows (CFD)** : Analyzing and modeling internal and external flows related to rotating machines, such as turbo-machinery and rotors. Experimental and numerical study of rotating machines (turbomachines, rotors, propellers) operating in degraded conditions, flow control, cavitation and multiphase flows are also topics of great interest. This subject gathers 6 researchers, 7 research engineers and about 13 PhD and post-doctoral students.
- **Flight dynamics in unsteady and non-uniform environments** : Development of analytical and experimental methods to determine the evolving behaviour of low-speed aircrafts in perturbed flight domains. Dynamic of post-stall behaviour, interaction with the environment and definition of experiments are subjects of interest. This topic gathers about 5 research engineers and 5 PhD and post-doctoral students.

1.2 Social and environmental responsibility

ENSAM's Corporate Social Responsibility (CSR) (or Responsabilité Sociétale des Entreprises (RSE)) action plan is structured around five strategic axes¹²:

1. **Strategy, governance and territorial anchoring**: presents ENSAM's actions in terms of sustainable development, in particular the Evolutive Learning Factories 4.0 project aimed at modernizing training and research platforms by fully integrating the CSR dimension.
2. **Education and training**: Training students in eco-responsibility
3. **Research and innovation**: Ensure that all research themes have direct or indirect impacts on the achievement of international and national sustainable development goals. Ensure free access to research results and data.
4. **Environmental management**: ENSAM aims to reduce its environmental impact by improving energy efficiency, cutting emissions (including Scope 3), and integrating carbon accounting. It also promotes biodiversity and sustainable mobility through staff engagement, responsible purchasing, and green-space management.
5. **Social policy**: Reconciling economic, societal, and environmental dimensions by training staff, teachers, research professors, and stakeholders in sustainable development issues, developing a quality of life policy within the institution, and ensuring equal opportunities in training, research, and human resources. An indicator is the QVT, that is detailed in the Rapport Social Unique (RSU)³ 2024.

The "Comité Social et Économique (CSE)" is the staff representative body within ENSAM. Although specific information on the CSE is limited to the 2022 RSU mentioning the monitoring of the actions of the "Comité d'Hygiène, de Sécurité et des Conditions de Travail National (CHSCTN)" which is the old name of the CSE.

¹<https://artsetmetiers.fr/fr/ecole-engagee>

²https://dp-www.s3.ensam.eu/public/2024-11/24_032_plaquette-bilan-RSE_V9_0.pdf

³https://dp-www.s3.ensam.eu/public/2025-07/RSU%202024_VCA.pdf

Arts et Métiers also publishes its activity reports, its social balance sheet / single social report and public data transmitted to the Commission des Titres d'Ingénieur (CTI) available online ⁴. More specifically, Students from Art et Métiers did a web page to learn about general RSE and sustainable development: ⁵ (addressing themes such as ecology, inclusion, gender equality and the fight against discrimination,...).

We can also find some documents about the evaluation of the LMFL⁶ on the Haut Conseil de l'Évaluation de la Recherche et de l'Enseignement Supérieur (HCERES) website.

My personal experience of the work environment is very positive. The working atmosphere within the laboratory is pleasant and conducive to research. The researchers are passionate about their work and share their knowledge with enthusiasm. The team is close-knit, which encourages the exchange of ideas and collaboration between members from all over the world. The working atmosphere suits me: the researchers generally work a lot, but there is no pressure on work schedules. The building is a beautiful old, typical Lille building, but well renovated and adapted to the research work being conducted (both numerical and experimental). It's nice to have space, interior courtyards, large amphitheatres, and a huge workshop for experiments.

⁴<https://artsetmetiers.fr/fr/observatoire-des-donnees>

⁵<https://artsetmetiers.fr/fr/actualites/developpement-durable-et-rse-faites-le-plein-de-connaissances-sur-savoir>

⁶https://www.hceres.fr/sites/default/files/media/publications/rapports_evaluations/pdf/

Chapter 2

Optimizing the performance of a centrifugal pump by data assimilation

The internship began in February 2025. I was provided with a fix computer, and credentials to connect to the Irene and Cassiopée supercomputers. Marcello MELDI, Antoine DAZIN and Miguel MARTINEZ VALERO which are my internship supervisors, showed me around the lab. Then, Tom MOUSSIE and Miguel, my co-offices, explained to me the basics of OpenFoam and CONES, the DA code. I was able to run a few tests on OpenFoam on the supercomputers, which allowed me to familiarize myself with the tools and the code.

2.1 Data Assimilation

DA[2] is a technique used to improve the accuracy of numerical models by combining them with observational data. It is widely used in various fields, including meteorology, oceanography, and engineering. The goal of data assimilation is to obtain a more accurate representation of the state of a system by integrating information from different physical measurements or high fidelity sources. An illustration of the data assimilation process is shown in Figure 2.1.

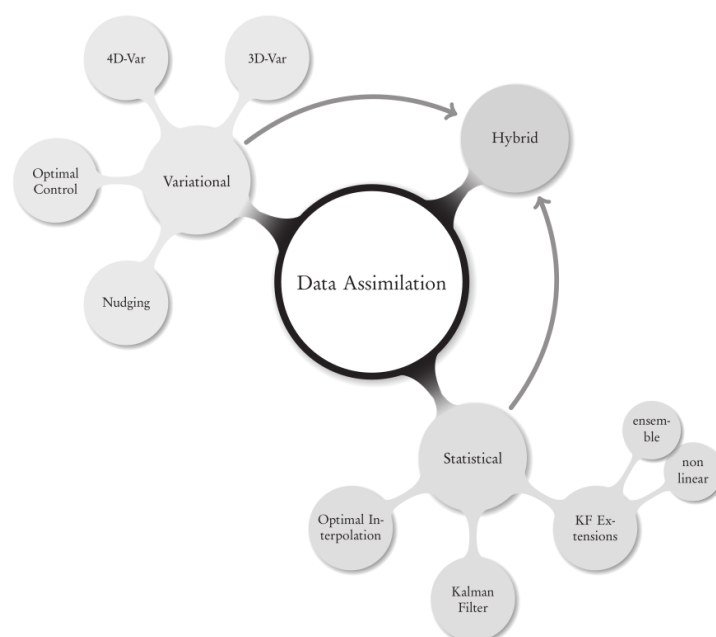


Figure 2.1: The big picture for DA methods and algorithms[2]

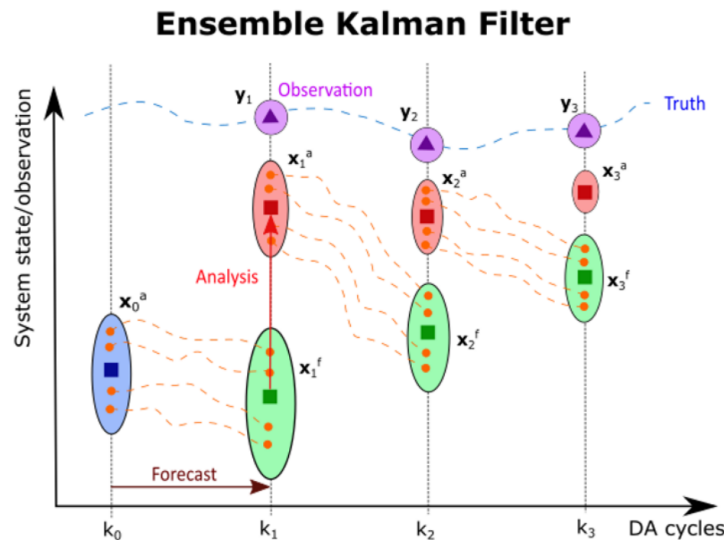


Figure 2.2: Illustration of the EnKF process (100% confidence in observations)

Expliquer théorème de Bayes etc... dans la partie maths ? (c'est ici ?) Expliquer plus en détail l'EnKF avec la fameuse figure.

2.1.1 State of the art

Our approach to data assimilation is based on the Ensemble Kalman Filter (EnKF), which is a Monte Carlo method that uses a set of model states (the ensemble) to estimate the state of the system. The EnKF is particularly well-suited for nonlinear systems and has been successfully applied in various fields, including computational fluid dynamics (CFD).

EnKF for DA in CFD has been used at LMFL since a few years, with the work of Lucas Villanueva[7], Miguel M.Valero[7], Tom Moussie[4], Sarp Er, Paolo Errante[4] and Marcello Meldi[7][4] and his colleagues. The EnKF method is a powerful tool for assimilating data into computational fluid dynamics (CFD) simulations, allowing for the correction of model predictions based on observed data.

Today, we can theoretically run a Direct Numerical Simulation (DNS) which resolves exactly the Navier-Stokes equations, but it is still too expensive to be used in practice. Therefore, we use Large Eddy Simulation (LES) or Reynolds-Averaged Navier-Stokes equations (RANS) models to simulate the flow. The EnKF method can be applied to these models to improve some of their parameters or physical state, to improve the physical accuracy.

Expliquer en quoi / cas test ou ça a fonctionné en CFD en utilisant des paramètres écoulement.
-> cas test cavity

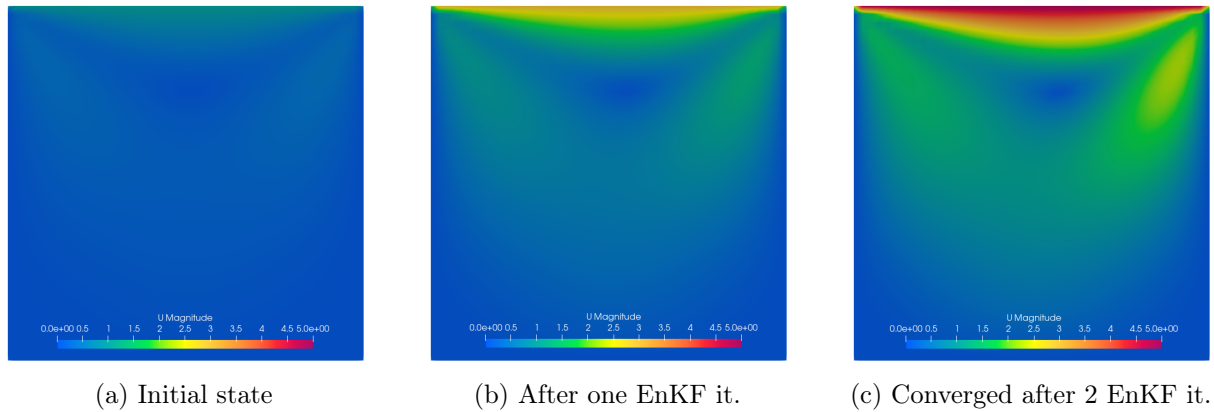


Figure 2.3: Evolution of the state through EnKF iterations

The IBM theory has been developed in 1972 by Charles S. Peskin[5]. The Immersed Boundary Method (IBM) is a numerical technique used to simulate fluid-structure interactions, particularly in cases where the structure is immersed in a fluid flow. It allows for the simulation of complex geometries and moving boundaries without the need for body-fitted meshes, which can be computationally expensive and difficult to implement.

IBM mais pas technique

(aller de + en plus en détail jusqu'à param IBM car gros cas test couteux en body-fitted)

Et là cas tests de thèse miguel (1 et 2)

Objectif : amélioration params forçage stationnaire

Centrifugal pumps are used since 1475[6] and are still widely used today in various applications, such as water supply, irrigation, and industrial processes. They are known for their efficiency and reliability, making them a popular choice for many fluid transport applications.



Figure 2.4: Old centrifugal pump used to move straw (personal picture)

2.1.2 Presentation of the test case: the centrifugal pump

The studied centrifugal pump is a SHF impeller, which is a type of centrifugal pump designed for high flow rates and low head applications.

Parler de la partie expérimentale

Results in the file are High-Speed Particle Image Velocimetry (PIV) data obtained by Antoine DAZIN in 2011[3] at a PIV frequency of 980 Hz on a SHF impeller. The operating conditions of the machine where :

Impeller characteristics		
R_1	Tip inlet radius	141.1 mm
R_2	Outlet radius	257.5 mm
Z	Number of blades	7
Q_d	Design flow rate	$0.236 \text{ m}^3/\text{s}$
K	Mean blade thickness	9 mm
Ω	Rotating velocity	1 200 rpm
β_2	Outside blade angle (from peripheral velocity $U = R_2\Omega$)	22.5°
Vaneless diffuser characteristics		
b_3	Diffuser width	38.5 mm
R_3	Inlet radius (without leakage, i.e., $R_3 = R_2$)	257.5 mm
R_4	Diffuser outlet radius	385.5 mm
t	Thickness of the upper and lower sections of the diffuser	20 mm

Tableau 2.1: Summary of the main geometrical and operating parameters of the impeller and diffuser at design conditions, employed in the numerical modelling of the centrifugal pump.

The data is stored in a directory structure that contains 6 directories

The 6 directories are corresponding to two tests realised at 25%, 50% et 75% of the diffuser height.

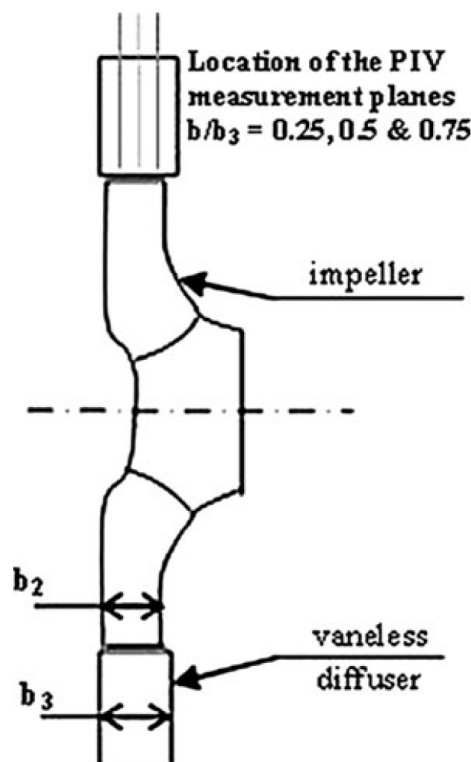


Figure 2.5: Position of the PIV plans

Each test was repeated twice for the same operating condition and the same height (R1, R2)

In each directory, there are 1581 files corresponding to instantaneous ASCII data file. The 9 columns are corresponding to :

x (m),y(m),Vx(m/s),Vy(m/s),r(m),theta(rad),Vr(m/s),Vtheta(m/s),Vz(m/s),Unknown

The pdf is a paper where can be found the experimental conditions even if the analysis are focussed on another flow rate.

The theoretical average speed at the input of the diffuseur is equal to

$$\frac{Q_{diffin}}{S_{diffin}} = \frac{Q_{pumpin}}{2\pi R_3 h_{diff}} = 3.65 m.s^{-1}$$

. $Q_{pumpin} = Q_{diffin}$ because the mass flow rate is constant and the flow is incompressible.

Détail : $R_3 = 0.257m$; $h=0.0400m$; $Q = 0.236m^3/s \rightarrow S=0.0646m^2$

Données hautes fidélités,...

2.2 Numerical ingredients

2.2.1 Governing equations

- Equations (Incompressible, turbulence,...) - Ajout IBM -> Methodo (forçage continue ici (vs interpolation ?)) -> avantages

Darcy's law in the body interface region Σ_b and the solid region Ω_b . To do so, a spatial dependent tensor $\mathbf{D}(\mathbf{x})$ must be defined. The resulting penalty term is modelled as follows:

$$\mathbf{f}_P = \begin{cases} \mathbf{0} & \text{if } \mathbf{x} \in \Omega_f \\ -\nu \mathbf{D}(\mathbf{u} - \mathbf{u}_{ib}) & \text{if } \mathbf{x} \in (\Sigma_b \cup \Omega_b) \end{cases} \quad (2.1)$$

$\mathbf{u}_{ib}(\mathbf{x}, t)$ is a target velocity representing the immersed body's physical behaviour. ... $\mathbf{u}_{ib} = \mathbf{0}$. The components D_{ij} of the tensor $\mathbf{D}$ are usually controlled to obtain the best compromise between the accuracy and stability of the numerical solver

$$Re_\Omega = \frac{R_2^2 \Omega}{\nu} = 5.53 \times 10^5$$

$$Ma = \frac{R_2 \Omega}{c} = 0.095 \text{ (incompressible)}$$

Reprendre ma présentation (dimensiosn de p)

$$\nabla \cdot (\mathbf{u}\mathbf{u}) = -\nabla p + \nu \nabla^2 \mathbf{u} - \overbrace{\frac{\xi}{\eta} (\mathbf{u} - \mathbf{u}_{ib})}^{\mathbf{f}_P}$$

$$\xi = \begin{cases} 0 & \text{if } \mathbf{x} \in \Omega_f \\ 1 & \text{if } \mathbf{x} \in (\Sigma_b \cup \Omega_b) \end{cases}$$

2.2.2 The open source code OpenFoam

To implemente the IBM forçage, we use the open source code OpenFoam, which is a C++ library for computational fluid dynamics (CFD). OpenFoam provides a wide range of tools for solving

the Navier-Stokes equations, including solvers, meshing tools, and post-processing tools. It is widely used in the CFD community and has a large user base.

More precisely, for example, to implemente the IBM forage on the speed U , we compute the forcing term f_P as follows:

```
1 // Compute the forcing term
2 volVectorField fP = -fP_ * (U - U_ib); EXEMPLE DE MISE EN FORME A CORRIGER
3 // Add the forcing term to the momentum equation
4 U += fP;
```

```
// Momentum predictor

MRF.correctBoundaryVelocity(U);

tmp<fvVectorMatrix> tUEqn
(
    fvm::div(phi, U)
    + MRF.DDt(U)
    + turbulence->divDevSigma(U)
    + cmptMultiply(D, U - UIB)
    ==
    fvModels.source(U)
);
fvVectorMatrix& UEqn = tUEqn.ref();

UEqn.relax();

fvConstraints.constrain(UEqn);
```

Figure 2.6: Model

To compute the time step Δ_t of the pimpleFoam simulation of the pump, we can use the Courant-Friedrichs-Lewy (CFL) condition, which states that:

$$\Delta_t = \frac{CFL \cdot \Delta_x}{\|U_\infty\|}, \quad (2.2)$$

where:

- CFL is the Courant number, typically set to 0.5 for stability. A way to visualize this is to think of it as the fraction of the distance a wave travels in one time step relative to the distance between two cells in the azimuthal direction at the tip of the impeller.
- Δ_x is the distance between two cells in the azimuthal direction at the tip of the impeller in our model, which is given as 1.5 mm or 0.0015 m.
- $\|U_\infty\|$ is the magnitude of the velocity at infinity, which can be approximated by the tangential velocity at the tip of the impeller.

$$\|U_\infty\| = \omega \cdot r_{tip} \quad (2.3)$$

where: - ω is the angular velocity in radians per second, - r is the radius at the tip of the impeller. Assuming the rotating speed is given in revolutions per minute (rpm), we can convert it to radians per second:

$$\omega = \frac{1200 \text{ rpm} \cdot 2\pi \text{ rad}}{60 \text{ s}} = 125.66 \text{ rps}. \quad (2.4)$$

Assuming the radius at the tip of the impeller is $r = 0.2566 \text{ m}$ (which is 0.2566m), we can compute the tangential velocity:

$$\|U_\infty\| = 125.66 \text{ rps} \cdot 0.2566 \text{ m} = 32.24 \text{ m/s}. \quad (2.5)$$

Assuming the distance between two cells in the azimuthal direction at the tip of the impeller is $\Delta_x = 0.0015$ m (which is 1.5 mm), we can now compute the time step Δ_t :

$$\Delta_t = \frac{0.5 \cdot 0.0015 \text{ m}}{32.24 \text{ m/s}} = 2.3 \times 10^{-5} \text{ s} \quad (2.6)$$

2.2.3 EnKF

Toujours mentionner les paramètres utilisés (dans le controlDict et le CONESDict)

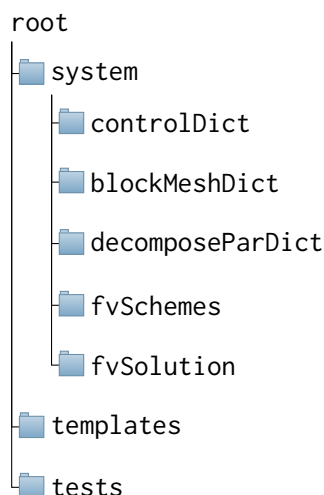
2.2.4 The library CONES

CONES is a library coded in Python/C++ developed for ... since 2022 (regarder sur le GitHub) ??? at LMFL It's parallelized and can be used on supercomputers. It is mainly designed to work with OpenFoam and DA codes as EnKF.

Simulations are performed using the C++ framework OpenFOAM, used to discretize the equations previously presented. Standing for *Open Source Field Operation and Manipulation*, OpenFOAM is an open-source project which provides many tools, referred to as executables

- **Pre-processing**, including meshing and decomposing tools.
- **Solving**, including in-built and custom solvers. Each solver is built to answer a specific CFD problem.
- **Post-processing**, including ParaView, an open-source visualization application.

OpenFOAM is not provided with a graphical interface, which is why a simulation is built by writing “dictionaries”, in which are stored the necessary information for the simulation (mesh, turbulence model, maximum number of characteristic times, ...). Thus, a specific directory structure must be observed, as outlined in ...



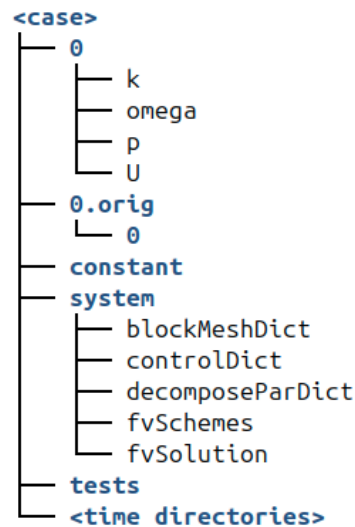


Figure 2.7: Arbo of OpenFoam

The **system** directory contains the main parameters concerning the progress of the simulation:

- The **controlDict** dictionary defines parameters such as start and end time, time steps, data output formatting, or data average computation.
- The **blockMeshDict** dictionary defines the geometry, mesh and boundary conditions implementation.
- The **decomposeParDict** dictionary defines the geometry and mesh decomposition, with the objective to run the simulation in parallel, using multiple cores.
- The **fvSchemes** dictionary defines the discretisation schemes used in the solution.
- The **fvSolution** dictionary defines the equation solvers, tolerances and other algorithm controls for instance.

The **constant** directory contains the different physical properties of the simulation. Moreover, the subfolder **polyMesh** contains the mesh data, obtained after the **blockMesh** command line is run:

- The **momentumTransport** dictionary defines the simulation type (RANS or LES), as well as the turbulence model.
- The **transportProperties** dictionary defines the transport model (Newtonian for instance) and some transport coefficients, such as the kinematic viscosity ν .
- The **turbulenceProperties** dictionary defines the different parameters of the chosen turbulence model.

The **0** directory contains the initial solution's fields. The **postProcessing** directory stores data from the running simulation, such as forces and probes' data. The **processor0...N** directories directly result from the geometry decomposition, which is proceeded through the **decomposeParDict** dictionary. The **dummy.foam** is an empty file which allows to visualize the simulation with ParaView.

2.2.5 Modeling

The objective of the model is to try to find same physical behavior as the experimental data. We call that digital twin.

For that we have first to discretize the numerical model of the pump. The mesh is generated using the `blockMesh` utility, which is a built-in OpenFOAM tool. The mesh is composed of 12 million cells. The mesh is composed of hexahedral cells. (quasi only, excepted some interface, cf figure). We don't need complex mesh because we don't need to fit the geometry of the impeller.

Boundary conditions are defined as that:

- Inlet: velocity inlet
- Outlet: pressure outlet
- Wall: no-slip wall

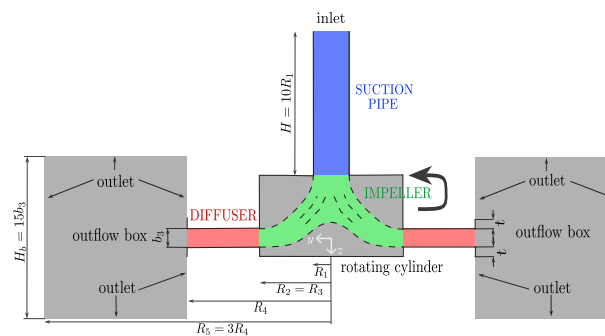


Figure 2.8: Model

2.3 IBM preliminary results

"Résultats de la simulation avec $3e5$ " -> $3e7$

2.4 Application de la DA

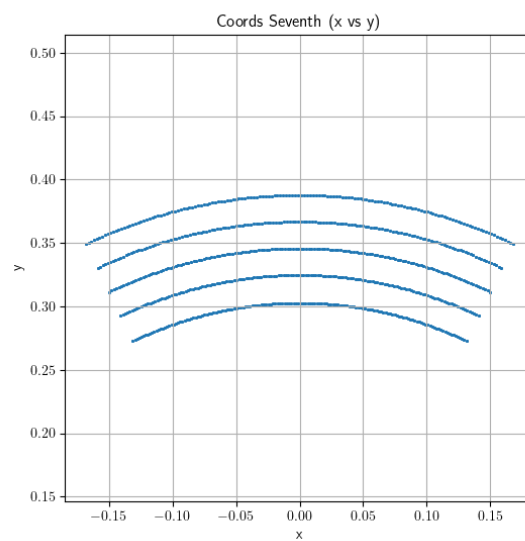


Figure 2.9: Interpolation points over 360/7 degrees

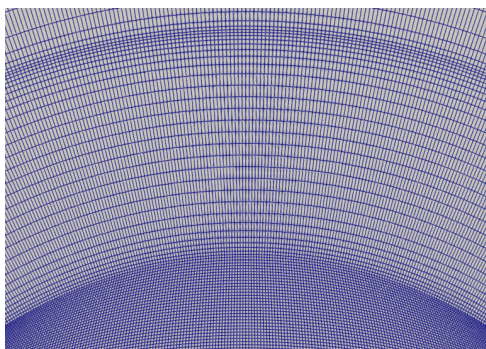


Figure 2.10: diffuseur Mesh

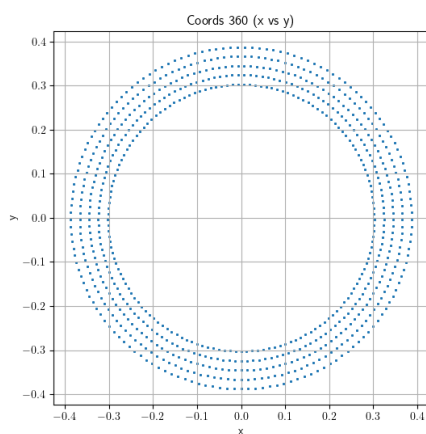


Figure 2.11: Interpolation points over 360 degrees

Explication de coords_seventh.txt, coords_360.txt et coords_7.txt

Tentative d'explication en français :

J'ai d'un côté une simulation numérique de la pompe centrifuge, qui est un modèle numérique de la pompe. De l'autre côté, j'ai des données expérimentales de PIV, qui sont des données expérimentales de la pompe centrifuge. Je veux utiliser les données expérimentales pour améliorer le modèle numérique de la pompe centrifuge. Pour cela, j'utilise la méthode de l'assimilation de données, qui est une méthode qui permet d'améliorer un modèle numérique en utilisant des données expérimentales. La méthode que j'utilise est l'Ensemble Kalman Filter (EnKF), qui est une méthode d'assimilation de données qui utilise un ensemble de modèles numériques pour améliorer le modèle numérique. Cela se fait en deux étapes :

1. Je fais une simulation numérique de la pompe centrifuge, en utilisant le modèle numérique de la pompe centrifuge. Cette simulation numérique me donne un ensemble de modèles numériques, qui sont les membres de l'ensemble.
2. Je compare les données expérimentales de PIV avec les données numériques de la simulation numérique. Pour cela, je fais une interpolation des données expérimentales de PIV sur les points de l'ensemble de modèles numériques. Cette interpolation me donne un ensemble de données expérimentales, qui sont les observations.

La comparaison entre les données expérimentales et les données numériques me permet de calculer l'écart entre les deux, qui est appelé l'innovation (mettre référence partie mathématique). L'innovation est utilisée pour mettre à jour les membres de l'ensemble de modèles numériques, en utilisant la méthode de l'EnKF. Cette mise à jour me permet d'améliorer le modèle numérique de la pompe centrifuge.

1. Ma simulation numérique de la pompe est stationnaire en utilisant la méthode MRF. Cela signifie que le champ de vitesse est constant dans le temps. Cependant, il n'est pas constant dans l'espace, car il dépend de la position dans le diffuseur. Un motif de vitesse répété sept fois est donc censé être visible (cela n'est pas vraiment le cas en instationnaire).

2. Les données PIV sont acquises à une vitesse de rotation de la pompe et une fréquence telle que 49 images sont acquises par tour de roue, soit 7 images entre deux passages d'aubes successives. Dans le référentiel de la pompe, les images sont prises à une différence de $360/49$ deg.

Nous voulons comparer des données consistantes. Nous pourrions synchroniser une simulation instationnaire avec les données PIV. Mais nous avons choisi de faire une simulation instationnaire dans un premier temps.

Nous pourrions reconstruire le champ spatial PIV (nous avons assez d'angle de vue et de données pour cela -> ref article Antoine). Cependant, la vitesse de rotation n'est pas très précise, mais surtout, la méthode MRF inverse la forme du sillage.

Nous avons donc choisi de moyenner temporellement les données PIV (ce qui revient à moyenner spatialement). Du côté de la simulation, nous avons donc besoin de moyenner azimutalement les données numériques pour les rendre compatibles avec les données PIV.

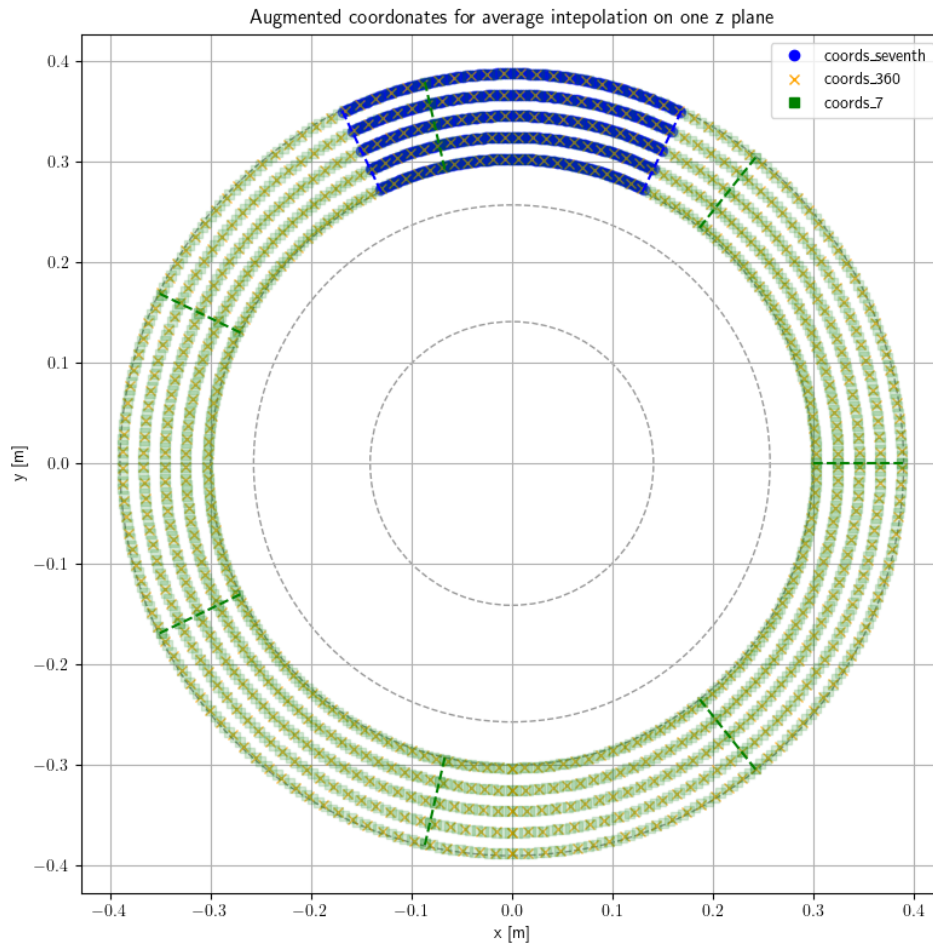


Figure 2.12: METTRE LA FIGURE AVEC LES PTS MOYENNES ROUGES ET VIOLETS

Imager avec des illustrations et vue en coupe ParaView

Pour cela j'ai commencé en moyennant $1/7$ mais les paramètres étant locaux (non globaux), j'ai ensuite moyenné sur 360 degrés, ce qui permet de mieux prendre en compte les variations locales des paramètres. Cependant, nous pouvons converger vers une moyenne physique qui atteints un minimum local mais qui ne respecte pas d'avoir 7 fois le même pattern. Cela permet à F_x d'être très différent de F_y mais aussi à la simulation de diverger avec des paramètres et vitesses erronées. Pour palier ce problème, je compare la moyenne des patternes sur mes 7 segments de $360/7$ degrés.

S'aider du plan de la présentation MALEAF + thèse M

Nous avons fait l'EnKF sans l'état... Sans inflation pour commencer. Si on converge vers une solution qui ne nous satisfait pas, on peut essayer d'ajouter... Cela permet de s'échapper d'un minimum local en ajoutant du bruit artificiellement.

Nous avons fait une simu RANS mais nous aurion pu faire une simu URANS (ou LES ou DDES) avec un modèle de turbulence...

L'expérience montre un ordre de grandeur de centaines d'itération entre les phases d'analyse, mais cela dépend beaucoup du cas. Il en est de même pour le nombre de phases d'analyse. Supposons 1000 itération en moyenne entre chaque phase d'analyse, et 300 phases d'analyse, cela nous fait 300 000 itérations à simuler, soit 93h, sur env. 2800 coeurs, soit 264kh.processeur de calcul.

Comment calculer la variance que je veux mettre ? Expérience : 5% De mon coté : je vise qu'en moyenne un membre d'ensemble ait un écart de $1e7$ par rapport au temps précédent (pour

des valeurs autour de $5e7$), donc, pour 30 membres d'ensemble, environ $95\% = 2 \text{ sigma}$ dans $5e7 \pm 1e7$, càd $\text{sigma} = 0.5e7 = 10\%$. Ensuite, le nombre d'itération entre les phases d'analyse est déterminé en regardant combien d'itération sont nécessaires pour faire varier la solution dans le diffuseur (influe, rapidement, converge doucement mais ce qui nous intéresse c'est que ça prenne la tendace) lorsqu'un paramètre de forçage est modifié (supposé très inférieure pour la turbulence car appliquée de partout)

The observation window was defined by observing the number of iterations required for the influence of the IBM to reach the diffuser and take the form of the converged result. We are not interested in the turbulence parameters to determine the observation window because they are present throughout the domain, their influence is almost direct (not necessarily their convergence). We chose 800 iterations

Pb de la DA : plus rapide que DNS mais toujours plus lent que RANS et le temps d'attente et de calcul sur supercalculateur (sur 4000 processeurs) est de l'ordre de plusieurs jours. On peut donc se demander si la DA est vraiment utile dans ce cas. En effet, la DA est plus rapide que la DNS mais pas que la RANS. Cependant, il faut prendre en compte le fait que la DA permet de réduire le temps de calcul de la simulation numérique en permettant d'optimiser les paramètres de l'IBM. De plus, la DA peut être utilisée pour d'autres cas d'étude où la DNS est trop coûteuse.

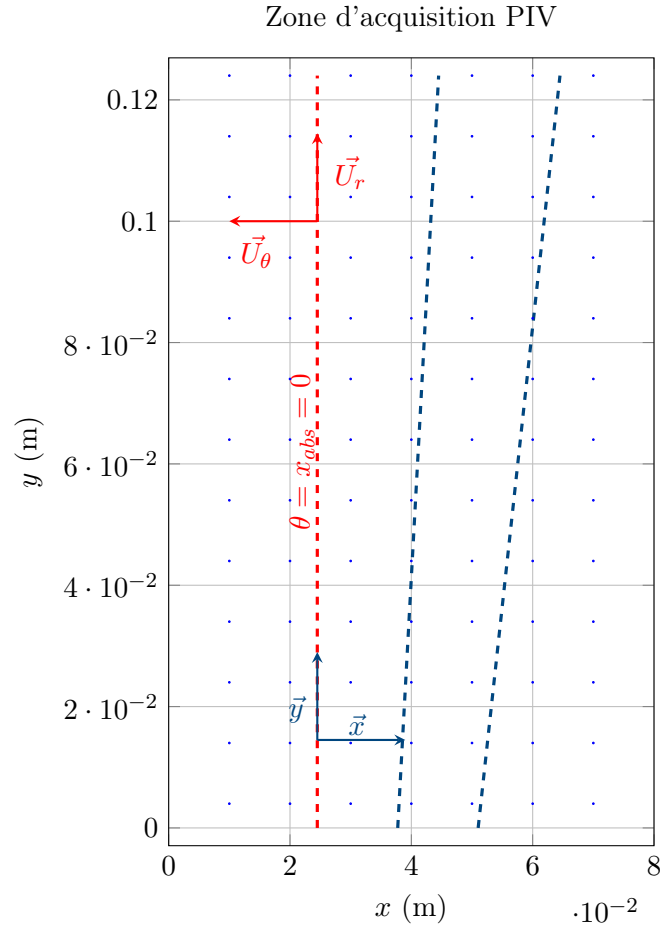
USER	ACCOUNT	BATCHID	NCPU	QUEUE	PRIORITY	STATE	RLIM	RUN/START	SUSP	OLD	NAME	MODES/REASON
thibault	gen1741	0982711	127	rene	278053	REN	24.00	-22h10	-	19.96	Pump3c7	Priority
thibault	gen1741	0982712	303	rene	269095	REN	2.00	-16h38	-	1.46	LCP18	Priority
thibault@rene183	test_renault_cv15	csc_app	-	-	-	-	-	-	-	-	-	-
USER	ACCOUNT	BATCHID	NCPU	QUEUE	PRIORITY	STATE	RLIM	RUN/START	SUSP	OLD	NAME	MODES/REASON
thibault	gen1741	0982711	127	rene	278053	REN	24.00	-22h10	-	19.96	Pump3c7	Priority
thibault	gen1741	0982712	303	rene	269095	REN	2.00	31.96	-	1.46	LCP18	Limit(2593,4096,4096,4096)

Figure 2.13: IRENE queue

2.4.1 Preprocessing of PIV data

Zone d'acquisition PIV

The PIV data is acquired in a rectangular zone of 0.08 m x 0.124 m. The cathesian coordinates are given in a local system, where the absolute origin is at the center of the pump.



Dans la simulation numérique, l'axe z est vers le centre de la Terre, contrairement à l'expérience où il est vers le haut. La roue tourne dans le sens trigonométrique dans le repère de la simulation numérique et dans le sens antitrigonométrique dans l'expérimentation. Actuellement, je respecte le formalisme de la simu num pour le repère.

Parler des coefficients de relaxation.

1. First, we need to know how to convert the coordinates from the local system to the absolute system. We find that between the line 55 and 56, the radius is constant and pass from lower than π to upper. In fact, for the plan at 75%, it is the same approach but between the line 65 and 66 because of some technical experimental constraints. We also did plot x over theta for three different values of theta and find out an intersection point for $x = 0.0245$. So now, we know that for $x = 0.0245$, $\theta = \frac{\pi}{2}$

$$\begin{cases} x_{rel} = x_{abs} + 0.0245 \\ y_{rel} = x_{abs} + R_2 \end{cases} \quad (2.7)$$

2.Average For now we are in stationary case (time invariant). So we are going to deal with data average over time. There are around 49 maps (=file) by turn. So to agerage the data over the time, we can average over the ϕ (ϕ appartient à 0-211rd) (=maps). That into the six directory.

3.Sampling We will also average data between R1 and R2 to go to 3 files (of 10x81x125 elements). Now, we keep only the data on the line $x = 0.0245$ (in fact we average data between x

= 0.024 and $x = 0.025$). Data are then truncate at line 89, where the radius is upper than 0.3, because of some laser reflexion on the pump during the PIV experiment.

4. Shaping for CONES Cones keep data in a particular way. We need to convert the data in a format that cones can read. From 3 files (of 10x81x125 elements), to the files `fieldCylindrical.txt` and `obs_coordinates.txt`. Describe shapes...

5. Creation of a file for the interpolation Finally, we need to create an `coords_seventh.txt` file for the interpolation function in CONES. We don't just use the coordinates of the points in the PIV data, because we want to average over the angle θ on the pump. The numerical simulation is $\frac{2\pi}{7}$ OU $2\pi/7$ periodic, so we interpolate in 150 points over theta between 0 and $2\pi/7$. To do that, we just take the `obs_coordinates.txt` file and duplicate 150 times the points over the angle θ (between $-\pi/7$ and $\pi/7$). In our case, we chose 150 that will approximately correspond to the shape of a cell at this radius. To be consistent with axis further, we take points between $-\pi/7 + \pi/2$ and $\pi/7 + \pi/2$

Here is the plot of the experimental data and the numerical data (for $D = 3e7$ (introduire la variable)), at the same points (both are averages over time and over the angle θ).

potentiellement en annexe

Concretely, to average the data over the angle θ , we need to modify the interpolation function in CONES. Instead of read coords, the interpolation function read the data in the `coords_seventh.txt` file.

Then, it computes the interpolation and average the data over the angle θ (between $-\pi/7$ and $\pi/7$), by doing a somme over θ and dividing by the number of points (150). To go back from 150x5x3x3 (to detail) to 5x3x3 data. All of those computation are mad in a cylindrical referencial. Ref au schéma tikz

A sixth coefficients DP has been add. It's represent the difference of pressure between the inlet of the pipe and the outlet of the diffuser. The numerical coefficient are interpolating, respectively at (0,0,-0.35) and (0,0.390,0). For the high fidelity value, we can find experimental data form the same pump [M. Fan 2022], that's match with M. Fan simulation (eq to approximately $\eta = 0.4$). So can also compute :

U_2 is the impeller tip speed computed from the impeller diameter and the rotation speed of the pump.

$$\frac{\Delta P_P}{\rho U_2^2} = \eta \Leftrightarrow \Delta P_P = \rho U_2^2 \eta \quad (2.8)$$

$$U_2^2 = (\omega * R_2)^2 = (2 * \pi * 1200 / 60 * 0.2566)^2 = 32,2 \text{ m/s}$$

As we are in a steady simulation, the result of the numerical pressure is divided by the density of the fluid.

$$\Delta P_P = 32.2^2 \times 0.4 = 415 Pa \quad (2.9)$$

So the state vector is now 46. Incertitude on P will be less to take more take it account. Inflation...

At pump flow rate = pump design flow rate So $Q^*/Q^* = 1$

2.4.2 Preparation/modifation on OpenFoam and CONES

Tout ce qu'il y a dans le CONES dict Parler des modifs dans le code ici.

- Réduction du pipe (avec recherche condition vitesse entrée)

2.5 Results

Ouverture : méthode IBM interpolation, article non référencé.

"Ouverture / idée" : DA temps réel dans avion impossible pour encore longtemps. Solution : Contrôle actif de la puissance en fonction des mesures de pression dans le compresseur : $f(\text{obs}) \rightarrow$ contrôle de la puissance Pour ça, se ref au sujet de thèse de Tarane obs pression $\rightarrow$ modèle de reconstruction du champ de vitesse $\rightarrow$ modèle de reconnaissance de patternes de décrochage $\rightarrow$ contrôle de la puissance Peut-être faire deux filtres de Kalman imbriqués ? hyperparamètres type fenetre observation, variance initiale, ...

Voir "TD $\rightarrow$ Done et "TDSave $\rightarrow$ Done" pour voir tt ce que j'ai fais

The random variations are random but with a reproducibility seed (sed) that allows to have the same result each time the simulation is launched.

Conclusion

Technical and human review

J'ai été enthousiasmé par ce stage au CERFACS, à plusieurs titres. Il m'a non seulement permis de développer et renforcer de nombreuses connaissances dans le domaine informatique mais également sur le plan humain. Il m'a donné une belle occasion de découvrir la vie d'un laboratoire de recherche.

L'objectif de ce stage était d'implémenter une méthode d'interpolation linéaire dans Antares puis de comparer sa précision avec la méthode IDW dans le cas de propagation acoustique avec l'analogie FWH, tout en optimisant la vitesse d'exécution du code. Après avoir exploré la structure d'Antares, une recherche documentaire a été menée pour savoir quelle méthode, implémentable dans la librairie, pouvait être candidate pour compléter la méthode IDW et linéaire. Quelques méthodes polynomiales et géostatistiques ont pu être mises en évidence. Une présentation un peu plus détaillée de la méthode par voisin le plus proche, IDW et linéaire a aussi été réalisée. L'un des principaux résultats de ce stage a été l'implémentation globalement réussie de la méthode d'interpolation linéaire, qui s'est révélée généralement plus précise que la méthode de Pondération Inverse à la Distance (IDW) précédemment codée. Coder l'interpolation linéaire jusqu'en 3D n'a été ni très difficile, ni chronophage. Cependant, l'intégrer au code déjà existant et faire les tests fut plus délicat. Le soutien de mon maître de stage a été primordial pour me montrer comment fonctionnent les outils au CERFACS et m'aider à sortir d'impasses. Cette nouvelle méthode a été testée sur des cas académiques mais aussi sur des cas industriels tels que sur une vue en coupe à la position $(0, 0, 0.01)$, selon l'axe z de la chambre de combustion preccinsta. L'étude sur les coefficients N et p de la méthode IDW a montré que pour des paramètres (N, p) autour de $(10, 10)$ nous pouvons obtenir de meilleurs résultats qu'avec la méthode linéaire, dans le cas d'aéroacoustique étudié. Ce travail a aussi amélioré les performances du code (cinq fois plus rapide pour la méthode IDW dans un cas test d'aéroacoustique). Le code ajouté pourrait aussi faciliter l'intégration de méthodes d'ordre supérieur grâce à certaines fonctions python réutilisables. J'ai ressenti tout au long de mon stage l'importance de réaliser un travail ouvrant des perspectives nouvelles pour un chercheur. Enfin, ce stage n'a pas été qu'une expérience professionnelle enrichissante. Il a conforté mon intérêt pour les méthodes numériques appliquées et la recherche en calcul scientifique. Je suis particulièrement reconnaissant pour l'accompagnement dont j'ai bénéficié tout au long de cette expérience et pour les opportunités de développement personnel et professionnel qu'il m'a offert.

Outlook

On pourrait également compléter les tests de la méthode linéaire sur la chaîne aéroacoustique, en observant la polaire de l'amplitude en fonction de l'observateur dans le cas d'un Mac non nul. Afin d'améliorer la rapidité, le code pourrait être passé en parallèle pour pouvoir s'exécuter sur plusieurs processeurs à la fois. Une petite amélioration qui permettrait de rendre le code compatible avec plus de bases serait d'inclure les bases "multizones" à l'interpolation linéaire.

Personal reflection

Le CERFACS est un laboratoire privé à la pointe des simulations numériques. Il s'est d'abord spécialisé dans l'aérodynamique et s'intéresse maintenant de plus en plus à la combustion. Il adapte ses domaines de recherche aux domaines importants de l'industrie. Durant ce stage, au-delà des aspects techniques, j'ai eu un bel aperçu du métier de chercheur tant au travers de mes missions que dans mes rencontres. Il s'agit de rechercher et de s'appuyer sur le travail déjà réalisé autour de notre problématique. Ensuite, il faut explorer des pistes qui peuvent s'avérer infructueuses après les avoir longuement explorées. Enfin, un travail de rédaction est nécessaire pour expliquer clairement notre travail aux scientifiques.

Annexes

Batch code used to run the simulation on the Rome partition of TGCC cluster.

```
#!/bin/bash                                # To be run with bash
#MSUB -r LCP35                             # Job name
#MSUB -o LCP35_%I.out                      # Output name. %I is the job ID
#MSUB -e LCP35_%I.err                      # Error name. %I is the job ID
#MSUB -q rome                             # Partition
#MSUB -A gen1741                           # Project ID
#MSUB -n 4446                             # Cores nb (35 members x 127 OF) + 1 python
##MSUB -c 2                               # To allocate 2 cores per task (default is 1)
##MSUB -N 1                               # Nb of calcul node (128 cores per node on rome)
#MSUB -T 43200                             # Elapsed time limit in seconds
##MSUB -Q long                             # (upto 3 days)
##MSUB -Q test                             # (short, high priority jobs)
#MSUB -m scratch                           # Execute in scratch directory
#MSUB -@ name@mail.fr:begin,end           # Email address

set -x
cd ${BRIDGE_MSUB_PWD}
module purge
module load gnu/8 mpi/openmpi/4 flavor/python3/cuda python3/3.10.6 openfoam/9
source $FOAM_SOURCE_FILE

./visuNParams.sh                          # Script .sh
ccc_mprun -f app_mySF35.conf              # Run the application with the configuration file
```

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List of acronyms

AI Artificial Intelligence

CERFACS Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique

CFD Computation Fluid Dynamics

CFL Courant-Friedrichs-Lewy

CHSCTN Comité d'Hygiène, de Sécurité et des Conditions de Travail National

CNRS Centre National de la Recherche Scientifique

CONES Coupling OpenFoam with Numerical EnvironmentS

CSE Comité Social et Économique

CSR Corporate Social Responsibility

CTI Commission des Titres d'Ingénieur

DA Data Assimilation

DNS Direct Numerical Simulation

EnKF Ensemble Kalman Filter

ENSAM École Nationale Supérieure d'Arts et Métiers

HCERES Haut Conseil de l'Évaluation de la Recherche et de l'Enseignement Supérieur

IBM Immersed Boundary Methods

IPSA Institut Polytechnique des Sciences Avancées

LES Large Eddy Simulation

LMFL Laboratoire de Mécanique des Fluides de Lille

ONERA Office National d'Études et de Recherche Aéronautiques

PIV Particle Image Velocimetry

RANS Reynolds-Averaged Navier–Stokes equations

RSE Responsabilité Sociétale des Entreprises

RSU Rapport Social Unique

Résumé L'objectif de ce stage était d'implémenter une méthode d'**interpolation linéaire** dans **Antares** puis de comparer sa précision avec la **méthode IDW** dans le cas de propagation acoustique avec l'analogie FWH, tout en optimisant la vitesse d'exécution du code. Premièrement, une recherche documentaire sur les méthodes d'interpolation implémentables dans notre cas a été menée. Ensuite, la méthode linéaire a été implémentée en utilisant une méthode particulière pour les cellules non triangulaires ou rectangulaires. L'efficacité a été augmentée dans les cas multi-instants partagés en évitant le recalcul des coefficients d'interpolation à chaque itération. Ces mêmes coefficients peuvent être récupérés par l'utilisateur qui voudrait appeler deux fois le traitement pour une base ayant une même structure. Le code est jusqu'à n fois plus rapide pour une base ayant n instants partagés. La méthode linéaire a été testée sur des cas simples et industriels. Elle fonctionne correctement sur tous les types de cellules, sauf dans certains cas très complexes comme des prismes ayant des faces opposées de tailles différentes et/ou non parallèles. Dans le cas de l'aéroacoustique, la méthode linéaire est plus précise que IDW dans tous les cas testés. Cependant, des résultats expérimentaux ont montré que des coefficients (N, p) autour de $(10, 10)$ donnaient de très bons résultats. Dans ce cas, IDW peut s'avérer plus précis que la méthode linéaire.

Abstract The objective of this internship was to implement a **linear interpolation** method in **Antares** and then to compare its accuracy with the **IDW method** in the case of acoustic propagation using the FWH analogy, while optimizing the code execution speed. First, a literature review on interpolation methods applicable to our case was conducted. Secondly, the linear method was implemented using a special method for non-triangular or rectangular cells. Efficiency was increased in shared multi-instant cases by avoiding the recalculation of interpolation coefficients at each iteration. These same coefficients can be reused by a user who wants to run the process twice on a base with the same structure. The code is almost n times faster for a database with n shared instants. The linear method has been tested on simple and industrial cases. It works correctly on all cell types, except in some very complex cases, such as prisms with non-parallel and/or differently sized opposing faces. In the case of aeroacoustics, the linear method is more accurate than IDW in all tested scenarios. However, experimental results have shown that coefficients (N, p) around $(10, 10)$ produce very good results. In such cases, IDW may prove to be more accurate than the linear method.